

Strange times ahead

THE LAST COUPLE OF WEEKS have seen some interesting announcements. Intel pulled out of the race to build mobile 5G modems right after Apple and Qualcomm resolved their differences. Their consumer processor roadmap leaked and shows that 14nm chips are going to be the mainstay for a lot more years. Also, the data centre wherein all major silicon companies enjoy constant growth is on a downturn. On Team Green's side, revenues had to be adjusted drastically since NVIDIA's RTX failed to convince both, gamers and investors alike. The data centre, professional visualisation, and OEM markets have slowed down. Gaming is still the largest contributor to NVIDIA's business. AMD on the other hand seems to be having a fun time as they bagged the rights to provide for Google Stadia. Their upcoming processor Zen 2 lineup promises 15% IPC gains and massive core counts at affordable prices. The graphics lineup, however, still has no promising leads over NVIDIA, but it might end up eating more into NVIDIA's GPU market share on the basis of affordability. Aside from all of these factors, IDC recently forecast a 7% global decline in semiconductor revenue.

If you look at all the three major companies, Intel has a diversified portfolio, AMD has some really interesting launches lined up and NVIDIA has just the consumer graphics portfolio to rely on. Considering the fact that AMD's RX 5000 series has potential and the fact that Intel is coming out with their discrete graphics platform, Xe in 2020, should have NVIDIA worried about losing more market share in the gaming business. Practically, all companies have at least one personal computing technology that would grow over the next year. Except NVIDIA, who must solely rely on the AI and autonomous business to continue growing if the consumer graphics market won't add to its already massive coffers. So while the announcements at major tech conferences and trade expos create the impression that all these companies have a lot going for them, underneath it all, there seems to be a worrisome trend as each company's strong suit is being affected negatively. All except for AMD, who's already been to the pits and has only one way to go, up. Its server chips have ridiculous core counts that allow for much higher density configurations but Intel has the advantage of VNNI which allows them to process AI work-

loads on the processors themselves rather than having to offload it onto a discrete accelerator. That's not saying that they aren't focused on discrete accelerators for they did announce a Nervana AI accelerator a while ago.

So what does this mean for consumers like you and I? Once the third-party benchmarks for each company's products come out, we'll get a clearer picture on who must pull up their socks and work on getting a better product out. If AMD's Zen 2 IPC improvement of 15% is indeed true, then it would be troublesome for Intel since that is where Intel has an advantage in the mainstream desktop space. Should AMD take the crown from Intel in single-threaded performance then we'll see a lot more entry-level and mid-range configurations use Ryzen processors exclusively. In markets such as India, Intel is already losing market share owing to reduced stock for their 9th Gen processors. A 9900K now costs 42K while a competing solution from AMD costs just 30K. In price sensitive markets such as ours, it makes little sense to spend the extra 12K to get a few extra frames in video games or similar performance gains in single-threaded applications. However, should Intel drastically revamp the IGP or come up with SKUs similar to Kabylake-G, then it'll be a game changer. The next couple of months are going to see so many different launches that will change the landscape of computing altogether. Pulling off way too many refreshes in a short period of time isn't exactly great for business, so there's going to be a lot of blood, but it's going to be great for us consumers. **d**



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